Design for Strength

Types of load! External forces grouped constitute what is called the load acting on the body. The nature of the load may vary from dead weight to varying load. There are classified as under

Static! A static or steady load is a stationary force, moment or torque acting on the member. It does not change its magnitude, point of application and direction

Impact or shock load! When a component is subjected to a sudden load or when the duration of loading is less than about half the natural period of vibration. Such loading is called impact or shock loading

Fatigue load! The alternating or repeated loading, whose duration is more than half but less than three times the natural period of vibration is called the fatigue load. In cyclic loading the load fluctuates between the maximum and minimum levels of loading. The loads may further be classified as

(a) Reversed
(b) Repeated
(c) Completely Reversed.

Failure criterion! The material behaviour varies from extreme brittleness to good ductility. The failure of a part made of brittle material may occur by fracture, while the ductile material may fail due to excessive yielding. Therefore, it is always important to know the cause of failure of mechanical component. The particular action subscribing to failure is termed criterion of failure. ~~Therefore~~

Theories of failure! The formal strategy which predicts the failure of a component is called the theory of failure. Some of them are

1. ~~The maximum pr~~ Maximum principal stress theory (Rankine's theory)
2. Maximum principal strain theory (Saint Venant' theory)
3. Maximum shear stress theory (Guest theory)
4. Maximum strain energy theory (Haigh's theory)
5. Maximum shear strain energy theory (Mises Henky theory)

(1) Maximum Principal stress theory! (Rankine theory) It is the simplest and the oldest theory of failure often called Rankine's theory after W.J.M. Rankine (1820-1872). According to this theory, permanent set takes place under a state of complex stress when the value of maximum principal stress is equal to that of yield point stress as found in a simple tensile test. or the minimum principal stress (the compressive stress) is equal to that of yield point stress in simple compression. Thus the condition of yielding are

$$\sigma_1 = f_y \quad \text{or} \quad |\sigma_3| = f_y' \quad \text{--- (1)}$$

f_y = yield point stress in simple tension
f_y' = — " ——————— simple compression

If the maximum principal stress is the design criterion the maximum principal stress must not exceed the working stress f for the material.

$$\boxed{\sigma_1 \leq f} \quad \text{--- (2)}$$

for brittle material which do not fail by yielding but fail by brittle fracture the maximum principal stress theory is considered to be reasonably satisfactory.

Maximum Principal Strain theory (St. Venant's theory)!-
According to this theory a ductile material begains to yield when the maximum principal strain reaches the strain at which yielding occur in simple test. or when the minimum principal strain (the compressive strain) equal the yield pt. strain in simple compression.

(a) In three dimensional stress system!

$$e_1 = \frac{\sigma_1}{E} - \frac{1}{mE}(\sigma_2+\sigma_3) \quad \text{--- (1)}$$

$$e_2 = \frac{\sigma_2}{E} - \frac{1}{mE}(\sigma_3+\sigma_1) \quad \text{--- (2)}$$

$$e_3 = \frac{\sigma_3}{E} - \frac{1}{mE}(\sigma_1+\sigma_2) \quad \text{--- (3)}$$

if e_y = yield pt. strain (tensile) $= f_y/E$

and e_y' = yield point strain (compressive) $= \frac{f_y'}{E}$

then according to this theory

$$e_1 = e_y \quad \text{—— (4)}$$

$$\sigma_1 - \frac{1}{m}(\sigma_2 + \sigma_3) = f_y \quad \text{—— (5)}$$

$$e_3 = e_y' \quad \text{—— (6)}$$

$$\left|\sigma_3 - \frac{1}{m}(\sigma_1 + \sigma_2)\right| = f_y' \quad \text{—— (7)}$$

Two dimensional stress system: Let σ_1 & σ_2 be the major and minor principal stresses then

$$\sigma_1 - \frac{1}{m}\sigma_2 = f_y \quad \text{—— (8)}$$

$$\sigma_3 - \frac{1}{m}\sigma_1 = f_y' \quad \text{—— (9)}$$

Thus the stress which, acting alone will produce same maximum strain e_1 is equal to $(\sigma_1 - \sigma_2/m)$. Hence the design criterion according to this theory will be

$$(\sigma_1 - \sigma_2/m) \leq f$$

f is working stress of the material.

This theory considers the intermediate principal stress σ_2, and is also satisfactory for brittle materials.

Maximum shear stress theory! (Guest theory): It predicts failure of a specimen subjected to any combination of loads when the maximum shearing stress at any point reaches the failure value (τ_f) equal to that developed at the yielding in an axial tensile or compressive test of the same material. Since the maximum shear is equal to half the difference between the maximum and minimum principal stress and since the maximum shear in simple tension is equal to half the tensile stress, we have

(4)

$$\tau_{max} = \frac{1}{2}(\sigma_1 - \sigma_2) = \frac{f_y}{2}$$

Hence the condition of yielding is

$$\sigma_1 - \sigma_3 = f_y$$

In case of 2D stress system, where σ_3 is zero the design criterion corresponding to an allowable stress f is

$$\sigma_1 - \sigma_2 = f$$

Maximum strain energy theory! (Haigh's theory)!

According to this a body under complex stresses fails when the total strain energy on the body is equal to the strain energy at elastic limit in simple tension.

(a) 3D stress system

$$U = \frac{1}{2E}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1}{m}\right]$$

where σ_1, σ_2 & σ_3 are of the same sign

Hence the yield criterion can be represented as

$$\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1}{m} = f_y^2$$

(b) 2D stress system: For 2D case $\sigma_3 = 0$

$$\sigma_1^2 + \sigma_2^2 - \frac{2\sigma_1\sigma_2}{m} = f_y^2$$

If f is working stress in the material, the design criterion may be

$$\left(\sigma_1^2 + \sigma_2^2 - \frac{2\sigma_1\sigma_2}{m}\right) < f^2$$

Maximum Shear strain energy (Mises-Henky):
(Distortion energy theory)

It state that inelastic action at any point in a body under any combination of stresses begin when the strain energy of distortion per unit volume absorbed at the pt. is equal to the strain energy of distortion absorbed per unit volume at any point in a bar stressed to the elastic limit under the state of uniaxial stress as occurs in a simple tension test.

The energy of distortion can be obtained by subtracting the energy of volumetric change from the total energy.

Total $U = U_v + U_s$ — strain energy of distortion — (1)

U_v — energy of volumetric change

$$U_s = U - U_v \quad \text{— (2)}$$

$$U_v = \frac{1}{2}(\text{average of } \sigma)\,\Delta V$$

$$U_v = \frac{1}{2}\left(\frac{\sigma_1+\sigma_2+\sigma_3}{3}\right)\Delta V$$

$$U_v = \frac{1}{2}\left(\frac{\sigma_1+\sigma_2+\sigma_3}{3}\right) \times \frac{1}{E}\left\{\sigma_1+\sigma_2+\sigma_3 - \frac{2}{m}(\sigma_1+\sigma_2+\sigma_3)\right\}$$

$$U_v = \frac{1}{2}\left(\frac{\sigma_1+\sigma_2+\sigma_3}{3}\right) \times \frac{1}{E}(\sigma_1+\sigma_2+\sigma_3) \times \left(\frac{m-2}{m}\right)$$

$$U_v = \frac{1}{2}\frac{(\sigma_1+\sigma_2+\sigma_3)^2}{3E} \times \left(\frac{m-2}{m}\right)$$

$$U_v = \frac{1}{6E}(\sigma_1+\sigma_2+\sigma_3)^2 \times \frac{(m-2)}{m}$$

$$U_v = \frac{m-2}{6mE}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2+2(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1)\right]$$

$$U_v = \frac{m-2}{6mE}(X+2Y) \quad \text{— (3)}$$

$X = \sigma_1^2+\sigma_2^2+\sigma_3^2$

$Y = \sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1$

$$U = \frac{1}{2E}\left[\sigma_1^2+\sigma_2^2+\sigma_3^2 - \frac{2}{m}(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_3\sigma_1)\right]$$

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$$U = \frac{1}{2E}\left[X - \frac{2}{m}Y\right] \quad \text{--- (11)}$$

Hence

$$U_s = U - U_v$$

$$U_s = \frac{1}{2E}\left[X - \frac{2}{m}Y\right] - \frac{m-2}{6mE}\left[X + 2Y\right]$$

$$U_s = \frac{1}{6E}\left[3X - \frac{6Y}{m} - X - 2Y + \frac{2X}{m} + \frac{4Y}{m}\right]$$

$$U_s = \frac{1}{3E}\left[X(1 + 1/m) - Y(1 + 1/m)\right]$$

$$U_s = \frac{(m+1)}{3mE}\left[X - Y\right]$$

$$U_s = \frac{m+1}{3mE}\left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1)\right]$$

$$U_s = \frac{m+1}{6mE}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]$$

This is the expression of strain energy of distortion.

<u>for 3D</u>

$$\frac{m+1}{6mE}\left[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2\right] = \frac{m+1}{3mE} f_y^2$$

<u>or</u>

$$\frac{1}{2}(\sigma_1-\sigma_2)^2 + \frac{1}{2}(\sigma_2-\sigma_3)^2 + \frac{1}{2}(\sigma_3-\sigma_1)^2 = f_y^2$$

<u>for 2D</u> $(\sigma_3 = 0)$

$$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 = f_y^2$$

If f is the working stress in the material

$$\boxed{\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 \geq f^2}$$

<u>Allowable Stress</u>: is the stress value which is used in design to determine the dimension of component

It is considered as a stress, which the designer expects will not be exceeded under normal operating condition. for ductile material the allowable stress σ is obtained by the following relationship.

$$\sigma = \frac{S_{yt}}{fs} \rightarrow \text{yield strength}$$

for brittle materials.

$$\sigma = \frac{S_{ut}}{fs} \text{ — ultimate tensile strength}$$

factor of safety:- while designing a component it is necessary to provide sufficient reserve strength in case of an accident. This is achieved by taking a suitable factor of safety (fs)

$$(fs) = \frac{\text{failure stress}}{\text{allowable stress}}$$

$$(fs) = \frac{\text{failure load}}{\text{working load}}$$

Stress Concentration! is defined as the localization of high stresses due to the irregularities present in the component and abrupt changes of the cross-section. In order to consider the effect of stress concentration and find out localized stresses a factor called stress concentration factor. It is denoted by K_t and defined as

$$K_t = \frac{\text{Highest value of actual stress near discontinuity}}{\text{Nominal stress obtained by elementary eq}^n\text{ for minimum cross-section}}$$

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$$K_t = \frac{\sigma_{max}}{\sigma_0} = \frac{\tau_{max}}{\tau_0}$$

σ_0 or τ_0 are stress determined by elementary eqn

σ_{max} or τ_{max} are localized stresses at the discontinuities.

fatique failure: is define as time delayed fracture under cyclic loading. For example transmission shaft, connecting road, gear, vehicle suspension springs and ball bearing.